

YOUSSEF ALI AICHA

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SUMMARY

Embedded systems engineer focused on designing custom PCB-based systems from firmware through software integration, with hands-on experience developing ESP32-S3 and ATmega328P platforms from schematic to deployment.

EDUCATION

IE UNIVERSITY

Bachelor of Science in Computer Science and Artificial Intelligence

Madrid, Spain

Expected: July 2029

- GPA 8.53/10 cumulative.
- Recipient of the 100% tuition UWC Scholarship and the Horizon Foundation Living Grant.

UNITED WORLD COLLEGE IN MOSTAR (UWCiM)

International Baccalaureate (IB) — Score: 36/45

Mostar, Bosnia and Herzegovina

Graduated: May 2025

SKILLS

- **Software:** C, C++, JavaScript, Git.
- **Embedded:** ESP32-S3, ATmega328P, Arduino, KiCad.
- **ML/Data:** Python, LightGBM, NumPy, Streamlit.
- **Languages:** Arabic (Native), English (Fluent), Turkish (Fluent), Spanish (A2).
- **Tools:** Git

TECHNICAL PROJECTS

AEGIS — Networked Rotating Ultrasonic Radar System

AEGIS

- Built a radar on an ESP32-S3: an HC-SR04 performs ultrasonic ranging to 300cm while a PWM-driven servo sweeps it in 2° increments, broadcasting telemetry over WebSocket at ~40Hz to a browser dashboard served entirely from the board's own flash.
- Designed a custom KiCad PCB and a non-blocking firmware architecture with runtime-adjustable sweep parameters and click-to-steer control. Full 180° sweep cycle in ~9s; 68% program storage utilization.

SENTINEL — Adaptive Traffic-Light Finite State Machine

SENTINEL

- Implemented a four-state traffic controller on bare ATmega328P with zero external libraries: polling loop gated entirely by millis() comparisons, no delay() calls, full 65s signal cycle, 2ms-multiplexed 7-segment display via 74HC595. 36% program storage utilization.
- Added a light-sensing night mode (ADC-sampled) that detects an idling vehicle and adaptively shortens the following phases, with UART serial diagnostics at 9600 baud. Backed by a custom KiCad PCB.

EXPERIENCE

Teknofest Electric Vehicle Initiative

Gaziantep, Turkey

Project Founder & Team Lead —

August - December 2021

- Led a 12-person student team through the design, budgeting, and prototyping of a full-scale electric vehicle chassis in CAD.
- Managed external sponsorships, placing within five points of the qualifying threshold for financial aid.

Institute of the Middle East

Aleppo, Syria

Project Communications and Data Officer

June - September 2025

- Coordinated communication pipelines and translated project documentation across Arabic, English, and Turkish for distributed regional development teams.

Designed a demand-driven placement strategy for EV charging infrastructure during a Datathon competition.

LEADERSHIP & ACTIVITIES

- DENEYAP Technology Scholar — completed 11 certified courses in design, robotics, electronics, and aeronautics as part of Turkey's first-generation elite cohort; State Honors for Best Eco-Friendly Robot.
- Founded the CS & Engineering Club (UWC Mostar) — restructured the club around student-led coding sessions.